

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 12040:1997

1997

READY MIXED CONCRETE

SPECIFICATION

ICS 91.100.30

BIS 1997

Indian Standard

1 SCOPE

This standard covers ready mixed concrete (RMC) produced in a central batching plant and delivered to the site in a truck mixer. It specifies requirements for production, delivery, and quality control of RMC.

2 DEFINITIONS

2.1 Ready Mixed Concrete

Concrete manufactured in a batching plant and delivered to the construction site in a truck mixer in a plastic (unhardened) condition.

3 GRADE DESIGNATION

RMC is designated by its characteristic compressive strength at 28 days in MPa. Common grades: M10, M15, M20, M25, M30, M35, M40, M45, M50.

4 MIX PROPORTIONS

Grade	Cement kg/m	Water L/m	Sand kg/m	Aggregate kg/m
M15	220	185	750	1100
M20	280	185	700	1080
M25	320	185	670	1060
M30	360	185	640	1040
M35	400	185	600	1020
M40	440	185	560	1000

5 QUALITY CONTROL

5.1 Slump Test

Concrete Class	Slump mm
Very Low (VL)	0-25
Low (L)	25-50
Medium (M)	50-100
High (H)	100-150
Very High (VH)	150-200

6 DELIVERY REQUIREMENTS

RMC shall be delivered to the site within 90 minutes of batching for normal conditions and within 60 minutes for hot weather conditions (ambient temperature >30°C). The truck mixer shall maintain a speed of 2-6 rpm during transport.